

Week 4: Key Concepts and Terms

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This deck collects the **key concepts and terms** from Week 4: Working with Data in Python.

For each term you should be able to:

- Define it in your own words
- Give an example or illustration
- Explain why it matters for your dissertation workflow

Use these slides as prompts — prepare your own definitions, answers and materials.

Note: This is a study aid only. It does not attempt to be comprehensive and should not be relied upon as sufficient preparation for assessment purposes.



Polars Data Structures

Term	What to know
DataFrame	What is a DataFrame? How is it like a spreadsheet with typed columns?
Series	What is a Series and how does it relate to a DataFrame column?
Data type	What are the common Polars types (Int64 , Float64 , String , Boolean , Date)? Why does every column have one?
null	What does null mean in Polars? How is it different from zero or NaN ?

Term	What to know
<code>.head(n) / .tail(n)</code>	What do these show and why are they useful for large datasets?
<code>.shape</code>	What does the tuple (<code>rows</code> , <code>columns</code>) tell you?
<code>.columns</code>	What does this return and when would you use it?
<code>.schema</code>	How does this differ from <code>.columns</code> ?
<code>.describe()</code>	What summary statistics does it produce? How is it like Excel's summary?



Reading and Writing Data

Term	What to know
<code>pl.read_csv()</code>	How do you load a CSV file into a DataFrame? What useful parameters does it have?
<code>.write_csv()</code>	How do you save a DataFrame to a CSV file?
<code>pl.read_excel()</code>	How do you read a specific sheet from an Excel file?
<code>.write_excel()</code>	What extra dependency is needed to write Excel files?
<code>try_parse_dates</code>	Why is this parameter especially useful for financial data?



Term	What to know
<code>pl.scan_csv()</code>	How does it differ from <code>read_csv</code> ? What does it return?
<code>LazyFrame</code>	What is a <code>LazyFrame</code> and why does deferring execution make queries faster?
<code>.collect()</code>	What does calling <code>.collect()</code> on a <code>LazyFrame</code> do?

Expressions and Contexts

Term	What to know
Expression	What is an expression in Polars? Why is it a “recipe” that does nothing on its own?
Context	What is a context? Why does Polars separate <i>what</i> to compute from <i>where</i> to put it?
<code>pl.col()</code>	What does <code>pl.col("price")</code> refer to? How do you use it in expressions?
<code>.alias()</code>	What does <code>.alias("new_name")</code> do and when do you need it?

Term	What to know
<code>select</code>	What does it return? How is it like choosing which columns to display?
<code>with_columns</code>	How does it differ from <code>select</code> ? Why use it to add a computed column?
<code>filter</code>	What kind of expression does it take? How is it like Excel's AutoFilter?
<code>group_by + agg</code>	What does grouping do? How is it like an Excel pivot table?



Working with Expressions

Term	What to know
Element-wise arithmetic	What does <code>pl.col("price") * pl.col("shares")</code> compute? Why are no loops needed?
Boolean column	What does <code>pl.col("price") > 200</code> produce? How do you combine conditions with <code>&</code> and <code> </code> ?
when / then / otherwise	How does this work like Excel's <code>IF()</code> ? Can you chain multiple conditions?
.cast()	What does casting do? What happens when you cast a float to an integer?

Term	What to know
Expression expansion	What does <code>pl.col("price", "shares").mean()</code> do? Why is it useful?
<code>.name.prefix()</code> / <code>.name.suffix()</code>	How do these help when applying the same operation to many columns?
Keyword syntax	What does <code>value = pl.col("price") * pl.col("shares")</code> mean inside <code>with_columns</code> ?

Working with Stock Returns

Term	What to know
<code>tidyfinance</code>	What does <code>tf.download_data()</code> do and what columns does it return?
Adjusted close	Why must you use adjusted close (not raw close) when computing returns? What does it correct for?
Stock split	What is a stock split and how does it affect the raw closing price?
Dividend adjustment	How do dividends affect stock prices, and why does adjusted close account for them?



Term	What to know
Daily return	What is the formula $r_t = p_t/p_{t-1} - 1$? What does it measure?
<code>.pct_change()</code>	What does this method compute? Why is the first row <code>null</code> ?
<code>.over("symbol")</code>	What is a window function? Why is <code>.over()</code> needed when you have multiple stocks?
Monthly returns	How do you compound daily returns into monthly? What does $(1 + r_1)(1 + r_2) \cdots (1 + r_n) - 1$ mean?
<code>.drop_nulls()</code>	When and why would you drop null values from your returns?



Combining DataFrames

Term	What to know
Join	What is a join? How is it like VLOOKUP for entire tables?
Join key	What is the shared column (e.g. ticker) that links two tables?
Inner join	Which rows survive? When would you use it?
Left join	Which rows survive? What happens when there is no match on the right?
Anti join	What does it return? Why is it useful for finding missing data?

Term	What to know
<code>pl.concat()</code>	What does concatenation do? When would you use it instead of a join?
Vertical concat	What does <code>how="vertical"</code> mean? What must be true about the columns?
Horizontal concat	What does <code>how="horizontal"</code> mean? What must be true about the rows?

Reshaping Data

Term	What to know
Wide format	What does a wide table look like? When is it useful (e.g. summary tables)?
Long format	What does a long (“tidy”) table look like? Why is it better for plotting and group_by ?
.pivot()	What does pivoting do (long → wide)? What are the index , on , and values parameters?
.unpivot()	What does unpivoting do (wide → long)? What new columns does it create?



Visualising Data

Term	What to know
<code>plotnine</code>	What is plotnine? Why do we convert to pandas with <code>.to_pandas()</code> before plotting?
Grammar of Graphics	What are the layers of a plot (data, aesthetics, geometry, theme)?
<code>ggplot() + aes()</code>	What does <code>aes()</code> do? How do you map columns to axes and colour?
<code>geom_*()</code> functions	What do <code>geom_line()</code> , <code>geom_col()</code> , <code>geom_histogram()</code> , and <code>geom_point()</code> each produce?
<code>labs() + theme_*()</code>	How do you add titles, axis labels, and visual polish to a plot?



Term	What to know
Price chart	How do you plot adjusted close over time using <code>geom_line()</code> ?
Return histogram	What does a histogram of daily returns show? What are “fat tails”?
Cumulative returns	What does <code>.cum_prod()</code> compute? How do you plot the growth of \$1?
<code>color="symbol"</code>	How does mapping colour to a grouping variable show multiple stocks on one chart?

Summary Checklist

Can you define and give an example for each?

- DataFrame / Series
- Data types (Int64, Float64, String)
- null
- .describe()
- read_csv / write_csv
- scan_csv / LazyFrame
- Expression / context
- select / with_columns
- filter / group_by
- when / then / otherwise
- .cast()
- .alias() / .name.prefix()
- Adjusted close
- .pct_change() / .over()
- Inner / left / anti join
- pl.concat()
- .pivot() / .unpivot()
- plotnine / Grammar of Graphics



Can you answer these in your own words?

- 1 What is a DataFrame and how does it differ from a Python list or dictionary?
- 2 What is the difference between `read_csv` and `scan_csv`?
- 3 What is the difference between `select` and `with_columns`?
- 4 Why do we use adjusted close instead of raw close when computing returns?
- 5 What does `.pct_change().over("symbol")` do, and why is `.over()` needed?
- 6 When would you use a left join vs an inner join?
- 7 When should data be in long format vs wide format?
- 8 In `plotnine`, what do `aes()` and `geom_line()` each do?

